

Pre-trained extreme value modelling

From amortised inference to reusable generative models

Edinburgh Summer School on Generative AI for Extremes

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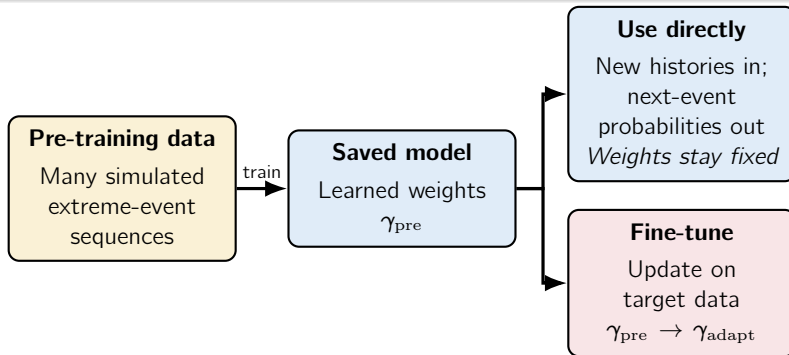


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What I mean by “pre-trained”

pretraining - Oxford English Dictionary

Computing. Training of a neural network or other AI system which sets the initial values of its parameters, after which further training with a different dataset can be performed to give the parameters their final values.



A pre-trained model: learn once, then reuse

- **Enables transfer learning:** Instead of training a new AI from scratch for every problem, a *pre-trained model* can be easily fine-tuned for specialised tasks;
- **Saves time and money:** Training an AI from scratch costs \$, while adapting an existing foundation model requires far less data and compute power.
- **Boosts accuracy and convergence time:** find a good starting point in the error surface, and then specialise.

Example: a transformer for cascading extremes

Pre-train on simulated token sequences; apply to financial histories directly, or fine-tune on financial training data.

The learned weights are reused. Fine-tuning is optional; performance on the target data still needs validation.

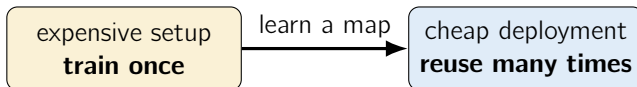
What I mean by “amortised”

Amortised - Oxford English Dictionary

the action or practice of gradually writing off the initial cost of an asset

- **asset:** an inference framework with a large training **cost** but negligible evaluation cost;
- The large computational cost of training is **amortised over time**, with the initial training cost **written off** thanks to repeated re-use of the **pre-trained** inference scheme.

What I mean by “amortised”



simulation, optimisation,
large training sets, GPUs

feed-forward evaluation,
fast estimation
or generation

The basic bargain

Spend substantial computation up front to construct a reusable function, then spread that cost over many future inferential or predictive tasks.

Framing follows Zammit-Mangion, Sainsbury-Dale and Huser (2025), *Annual Review of Statistics and Its Application*.

Why should we care for extremes?

Extreme-value models are often expensive

- high-dimensional likelihoods can be unavailable or intractable;
- censoring and partial derivatives/integrals add cost;
- dependence models can be expensive to fit repeatedly;
- rare-event simulation may itself be demanding.

But the same task is often repeated

- many spatial windows / sites / times;
- online or operational updating;
- threshold and design sensitivity analyses;
- scenario simulation under many conditions;
- repeated prediction of evolving cascades.

Expensive + repeated = a natural target for amortisation.

Motivating example: max-stable processes

Max-stable processes (MSPs), which arise as the **only possible non-degenerate limit of pointwise maxima of i.i.d random fields**, are popular models for spatial extremal dependence.

A MSP with unit Fréchet margins has the construction

$$Z(\mathbf{s}) = \sup_{k \geq 1} R_k W_k(\mathbf{s}),$$

where $\{R_k\}_{k \in \mathbb{N}}$ are points of a Poisson process on $(0, \infty)$ with intensity $r^{-2}dr$ and $\{W_k(\mathbf{s})\}_{k \in \mathbb{N}}$ are i.i.d. copies of a non-negative stochastic process $W(\cdot)$ satisfying $\mathbb{E}[W(\mathbf{s})] = 1$ for all $\mathbf{s} \in \mathcal{S}$.

Motivating example: max-stable processes

For observations $\mathbf{z} = (z_1, \dots, z_D)^\top$ from a max-stable process with unit Fréchet margins,

$$\Pr\{Z(\mathbf{s}_1) \leq z_1, \dots, Z(\mathbf{s}_D) \leq z_D\} = \exp\{-V(\mathbf{z}; \boldsymbol{\theta})\}, \quad z_1, \dots, z_D > 0.$$

Assuming the required mixed derivatives exist, the joint density is

$$f(\mathbf{z}; \boldsymbol{\theta}) = \exp\{-V(\mathbf{z}; \boldsymbol{\theta})\} \sum_{\pi \in \mathcal{P}_D} \prod_{B \in \pi} \{-V_B(\mathbf{z}; \boldsymbol{\theta})\},$$

where $\mathcal{P}_D$ is the collection of all partitions of $\{1, \dots, D\}$ and

$$V_B(\mathbf{z}; \boldsymbol{\theta}) = \frac{\partial^{|B|} V(\mathbf{z}; \boldsymbol{\theta})}{\prod_{j \in B} \partial z_j}.$$

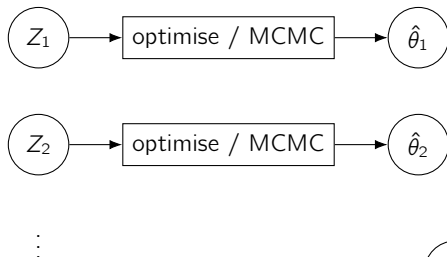
Motivating example: max-stable processes

- Number of terms in the likelihood grows **faster-than-exponentially**
- D -th Bell number: $D = 3 \Rightarrow 5$; $D = 10 \Rightarrow 115975$;
- **Computational tractability** of the likelihood is limited (generally) to $D \leq 12$ (Castruccio et al., 2016);
- A lot of time has been spent on researching efficient likelihood-based inference techniques for MSPs, e.g., via **pairwise likelihoods**;
- Computational issues are **compounded by (left) censoring (we will come back to this later...)**

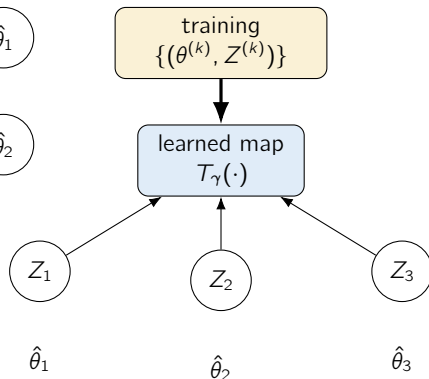
Castruccio, S., Huser, R., and Genton, M. G. (2016). *JCGS* 25.4: 1212-1229.

Non-amortised versus amortised computation

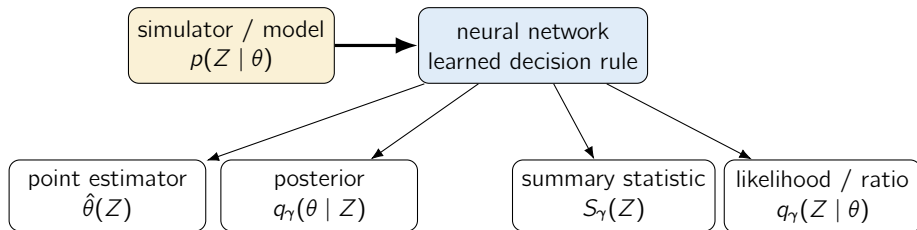
Non-amortised



Amortised



What can be amortised? A recent review



Where is amortised inference already being used?

Target	Example domains	Why amortisation matters
Point estimation	satellite CO ₂ retrieval; spatial extremes	very high throughput; huge numbers of local fits
Approximate posterior	neuroscience; population genetics; geophysical inverse problems; cognitive response times	posterior inference for expensive simulators or very large data collections
Neural likelihood	biophysical neuron models; stochastic differential-equation models	replace repeatedly expensive or unavailable likelihood evaluations
Likelihood ratio	particle physics; gravitational waves; cosmology	rapid parameter inference in simulator-driven science

One extremes example

Richards et al. (JMLR; 2024) use neural Bayes estimators to fit hundreds of thousands of spatial-extremes models to censored particulate-matter data.

Relevant literature

1 Neural point estimation:

- Lenzi, A., Bessac, J., Rudi, J., & Stein, M. L. (2023). Neural networks for parameter estimation in intractable models. *Computational Statistics & Data Analysis*, 185, 107762.
- Sainsbury-Dale, Matthew, Andrew Zammit-Mangion, and Raphaël Huser. Likelihood-free parameter estimation with neural Bayes estimators. *The American Statistician* 78.1 (2024): 1-14.

2 Likelihood approximation:

- Walchessen, J., Lenzi, A., & Kuusela, M. (2024). Neural likelihood surfaces for spatial processes with computationally intensive or intractable likelihoods. *Spatial Statistics*, 62, 100848.
- Majumder, R., Reich, B. J., & Shaby, B. A. (2024). Modeling extremal streamflow using deep learning approximations and a flexible spatial process. *The Annals of Applied Statistics*, 18(2), 1519-1542.

Relevant literature

1 Neural classification:

- André, L. M., Wadsworth, J. L., & Huser, R. (2026). Neural Bayes estimation and selection of complex bivariate extremal dependence models. *Extremes*, 29(2), 347–386.
- Wixson, T.P., Cooley, D. Neural classification of asymptotic (in)dependence. *Extremes* 29, 387–417 (2026).

2 Neural posterior inference:

- Pasquier, Carlos A., and Luis A. Barboza. Amortized Bayesian Inference for Spatio-Temporal Extremes: A Copula Factor Model with Autoregression. *arXiv:2510.02618* (2025).
- Maceda, Hector, Lenzi, & Reich. (2026). Demonstrating the power and flexibility of variational assumptions for amortized neural posterior estimation in environmental applications. *Environmetrics*, 37(1), e70074.

3 Neural amortised simulator: Walchessen, J., Zammit-Mangion, A., Huser, R., & Kuusela, M. (2025). Neural conditional simulation for complex spatial processes. *arXiv:2508.20067*.

Amortised inference versus amortised models

Amortised inference

$$Z \xrightarrow{I_\gamma} \text{inferential target}$$

- a statistical model / simulator is specified;
- learn to infer its parameters or posterior quickly;
- examples: NBE, neural posterior, neural likelihood ratio.

Amortised model

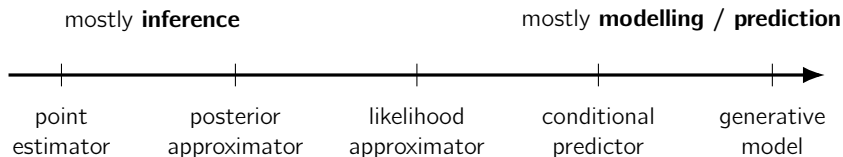
$$(c, \varepsilon) \xrightarrow{G_\gamma} Y$$

- the learned network is itself a reusable distributional / generative mechanism $P(\tilde{Y}|c)$;
- condition on covariates, history, or scenario - encapsulated in c ;
- generate or predict without fitting a new model each time.

Same computational philosophy; different object is being amortised.

"Amortised model" is used here as lecture terminology to distinguish a reusable learned generative law from a reusable inferential rule.

It is better viewed as a spectrum



The common mathematical pattern is an expected-risk problem over a distribution of tasks:

$$\gamma^* = \arg \min_{\gamma} \mathbb{E}_{\mathcal{T}}[\mathcal{L}\{\mathcal{T}, F_{\gamma}(\mathcal{T})\}].$$

The crucial design question is therefore: **what is the task distribution over which we want to generalise?**

Lecture roadmap

- ① **Amortised inference:** why and when?
- ② **Neural Bayes estimators:** likelihood-free point estimation and uncertainty quantification for extremes. Joint with Sainsbury-Dale, Zammit-Mangion, Huser.
- ③ **Amortised generative models:** a transformer for cascading extremes. Joint with Lee, de Carvalho, Sabanis.

The theme throughout: **learn once over a deliberately chosen family of tasks, then reuse.**

Neural Bayes estimators

Neural point estimators

A **neural point estimator** is a neural network

$$\hat{\theta}_{\gamma}(\mathbf{Z}) : \mathcal{Z} \rightarrow \Theta$$

that maps an observed dataset directly to a parameter estimate.

Training recipe

- 1 draw $\theta^{(k)} \sim \Omega(\cdot)$ from a prior / training distribution;
- 2 simulate $\mathbf{Z}^{(k)} \sim p(\cdot | \theta^{(k)})$;
- 3 minimise a loss $L\{\theta^{(k)}, \hat{\theta}_{\gamma}(\mathbf{Z}^{(k)})\}$ over many simulated pairs.

No likelihood evaluation is required during deployment: only a feed-forward operation.

Why call it a *Bayes* estimator?

Connecting neural estimators to classical estimators?

- A non-negative **loss function**, $L(\boldsymbol{\theta}, \hat{\boldsymbol{\theta}}(\mathbf{Z}))$, assesses an estimator, $\hat{\boldsymbol{\theta}}(\cdot)$, for a single parameter vector, $\boldsymbol{\theta}$, and model realisation, $\mathbf{Z}$.
- The **Bayes risk** averages the loss function over the sample space, $\mathcal{S}$, and the parameter space, Θ , with respect to the prior, $\Omega(\cdot)$;

$$r_{\Omega}(\hat{\boldsymbol{\theta}}(\cdot)) = \int_{\Theta} \left[\int_{\mathcal{S}} L(\boldsymbol{\theta}, \hat{\boldsymbol{\theta}}(\mathbf{z})) f(\mathbf{z} | \boldsymbol{\theta}) d\mathbf{z} \right] d\Omega(\boldsymbol{\theta}),$$

where $f(\mathbf{z} | \boldsymbol{\theta})$ is the probability density function of the data conditional on $\boldsymbol{\theta}$.

- A **minimiser of the Bayes risk** is said to be a **Bayes estimator** with respect to $L(\cdot, \cdot)$ and $\Omega(\cdot)$.

Neural Bayes estimators

- Denote a neural estimator by $\hat{\theta}_{\boldsymbol{\gamma}}(\cdot)$, where $\boldsymbol{\gamma}$ is a vector of neural-network parameters (“weights” and “biases”).
- A neural estimator is trained by solving the optimisation task,

$$\boldsymbol{\gamma}^* = \arg \min_{\boldsymbol{\gamma}} \frac{1}{K} \sum_{k=1}^K L(\boldsymbol{\theta}^{(k)}, \hat{\theta}_{\boldsymbol{\gamma}}(\mathbf{z}^{(k)})), \quad (1)$$

where $\boldsymbol{\theta}^{(k)}$, $k = 1, \dots, K$, is sampled from the prior $\Omega(\cdot)$ and, for each k , data $\mathbf{z}^{(k)}$ are sampled from $f(\cdot \mid \boldsymbol{\theta}^{(k)})$.

- Since the objective function in (1) is a Monte Carlo approximation of the Bayes risk, **neural estimators approximate the Bayes estimator**.

Sainsbury-Dale et al. (2024), The American Statistician 78(1), 1—14.

The loss chooses the inferential target

Loss	Bayes target	Neural output
$(\hat{\theta} - \theta)^2$	posterior mean	$\mathbb{E}(\theta \mid Z)$
$ \hat{\theta} - \theta $	posterior median	$\text{med}(\theta \mid Z)$
quantile / tilted loss	posterior q -quantile	$Q_q(\theta \mid Z)$
classification loss	posterior model probability	$P(M = m \mid Z)$

Useful consequence

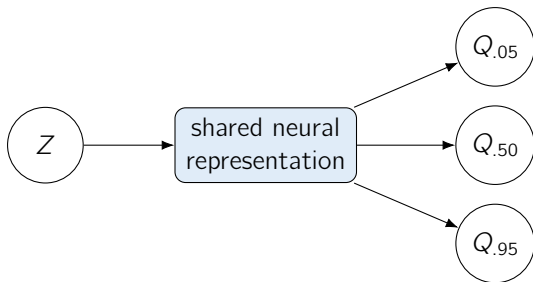
We can train separate or joint outputs for posterior quantiles and obtain rapid marginal uncertainty quantification without running a bootstrap or MCMC for every new dataset.

Amortised uncertainty quantification

For scalar θ , one form of the quantile loss is

$$L_q(\theta, \hat{\theta}) = (\hat{\theta} - \theta) \{\mathbb{I}(\hat{\theta} > \theta) - q\}.$$

Its Bayes estimator is the posterior q -quantile.



Architectural constraints can enforce $Q_{.05} \leq Q_{.50} \leq Q_{.95}$ and prevent quantile crossing. Sainsbury-Dale, Zammit-Mangion, Richards, and Huser (2025), JCGS.

The architecture should respect the data

If $Z_1, \dots, Z_m$ are conditionally independent replicates given θ , a Bayes estimator is permutation invariant:

$$\hat{\theta}_{\text{Bayes}}(Z_1, \dots, Z_m) = \hat{\theta}_{\text{Bayes}}(Z_{\pi(1)}, \dots, Z_{\pi(m)}).$$

A natural neural architecture is a **DeepSet**:

$$\hat{\theta}(Z_1, \dots, Z_m) = \phi[a\{\psi(Z_1), \dots, \psi(Z_m)\}],$$

where $a(\cdot)$ is a permutation-invariant aggregation such as a sum or mean.

Principle

Do not ask a generic network to relearn symmetries or invariances that the statistical problem already tells us are present.

Different data structures, different networks

Data structure	Possible architecture	Statistical structure encoded
vectors / summaries	MLP	none beyond dimensions
regular spatial grids	CNN	local spatial translation structure
irregular spatial data	graph neural network	neighbourhood graph, arbitrary sites
replicated datasets	DeepSets	permutation invariance
time-series / sequences	recurrent net / transformer	ordering and temporal dependence
censored / missing data	masks + structured network	which values are observed / censored

Why spatial extremes are a good NBE test case

Classical bottleneck

For popular spatial extreme-value models, full likelihood evaluation can become prohibitive as dimension grows. Peaks-over-threshold likelihoods additionally involve censoring contributions and multivariate integrals.

Simulation remains feasible

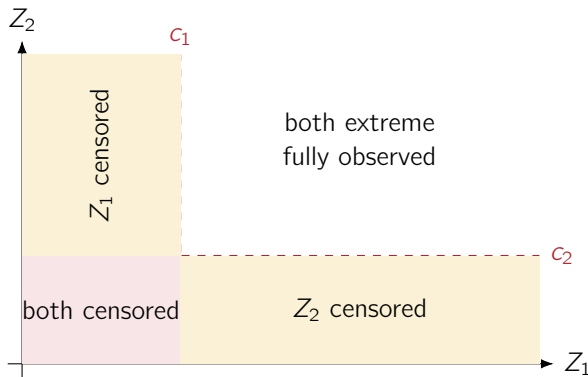
Many models are straightforward to simulate from conditional on their parameters, exactly the regime where simulation-based amortised estimators are useful.

Likelihood hard; simulator available; many fits required.

Richards et al. (2024), JMLR 25(390):1–49; Huser and Wadsworth (2022).

Censoring: a problem for the likelihood and the network

In peaks-over-threshold inference, non-extreme components may be deliberately treated as left-censored.



The exact values of censored components should not influence the estimator, but the censoring pattern should.

Encoding censored extremes for an NBE

Practical construction - supply both transformed data and a censoring indicator:

$$(Z^*, \mathcal{I}), \quad \mathcal{I}_j = \mathbb{I}\{Z_j < c_j\}, \quad \Pr\{\mathcal{I}_j = 1\} = 1 - \tau.$$

- 1 transform observations to standard margins with a finite lower endpoint;
- 2 replace censored values by a constant c^* outside the support;
- 3 include binary mask $\mathcal{I}$ so the network knows which components were censored;
- 4 train entirely on data processed with the same censoring rule.

Interpretation

The input representation intentionally deletes information that the censored likelihood would integrate out.

Amortise over nuisance choices too

Suppose the desired censoring threshold is indexed by τ .

Narrow training distribution

Train at one fixed $\tau = 0.9 \Rightarrow$ the deployed estimator is specialised to that threshold.

Broader task distribution

Randomise τ during training and provide it as an input:

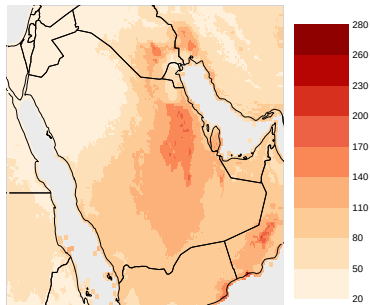
$$\tau^{(k)} \sim \text{Unif}(0.85, 0.95), \quad \hat{\theta}_{\gamma} = \hat{\theta}_{\gamma}(Z^*, \mathcal{I}, \tau).$$

Then a new user can change τ without retraining from scratch.

Amortisation is only as broad as the task distribution used during training.

Application

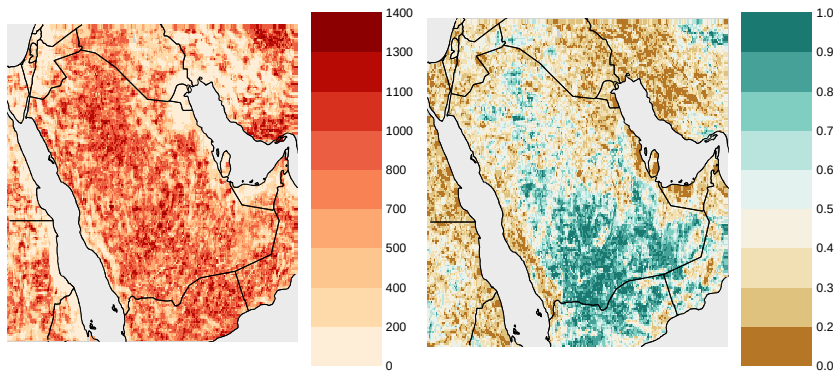
Application to monthly Saudi Arabian $\text{PM}_{2.5}$ (Van Donkelaar et al., 2021) concentrations, 1998–2020, shows the computational gains of our **amortised estimator**.



Observation of **surface average $\text{PM}_{2.5}$ conc. ($\mu\text{g}/\text{m}^3$)** for Jul. 2012.

Results

Each pixel is a **single model fit**.



A concrete pay-off: repeated spatial-extremes fitting

- Monthly Saudi Arabian PM_{2.5} fields.
- Local anisotropic spatial-extremes models fitted over many moving windows.
- Five amortised estimators smoothing/grid resolutions.
- More than 130 million model fits in the analysis.
- Computational speed-up of OOM 10 million.

Training

roughly 24–72 hours once

Deployment

roughly 1–4 milliseconds per fit

Scale

up to spatial dimension $32^2 = 1024$

The point is not just a faster fit; it makes a previously unrealistic *analysis design* feasible.

Amortised estimators – a tool for everyone

e3-10 *Raphaël Huser, Thomas Opitz and Jennifer L. Wadsworth*

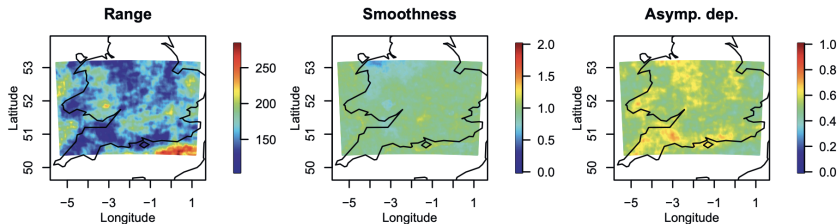


Figure 4. Three of the five estimated parameters obtained from local fits of the anisotropic Huser and Wadsworth (2019) copula model in all subgrids of size 32×32 ($d = 1024$) within the study domain for the UKCP precipitation example from Figure 1. The plots display, for each local subgrid, the estimated range parameter (left), smoothness parameter (middle), and shape parameter (right) controlling the asymptotic dependence type; the two anisotropy parameters (stretch and rotation) are not shown for brevity. All of these estimates were obtained in a few minutes in total by reusing the pretrained censored neural Bayes estimators from Richards et al. (2024).

Huser R, Opitz T, Wadsworth JL. Environmental Data Science. 2025;4:e3.

What an NBE learns – and what it does not

Learned

- a fast approximation to a Bayes decision rule;
- nonlinear summaries of the data;
- how those summaries map to the chosen posterior functional.

Still specified by us

- model / simulator family;
- prior or parameter training distribution;
- loss and inferential target;
- architecture and invariances;
- censoring / missingness mechanism used in training.

A neural Bayes estimator is **not** a licence to stop thinking statistically.

Software and usage

- An R interface to a Julia package, **NeuralEstimators**, is available online¹ with censored inference also illustrated²
- Sainsbury-Dale et al. (2025). Neural Bayes estimators for irregular spatial data using graph neural networks. JCGS, 34(3), 1153-1168.
- Lee et al. (2026). Spatial Extremes at Scale: A Case Study of Surface Skin Temperature and Heat Risk in the United States. arXiv:2604.18840.
- Sainsbury-Dale et al. (2025). Neural parameter estimation with incomplete data. arXiv:2501.04330.
- Rödder, A., Hentschel, M., & Engelke, S. (2025). Theoretical guarantees for neural estimators in parametric statistics. arXiv:2506.18508.

¹<https://msainsburydale.github.io/NeuralEstimators.jl/dev/>

²<https://github.com/Jbrich95/CensoredNeuralEstimators>

Software and usage

- André et al. (2026). Neural Bayes estimation and selection of complex bivariate extremal dependence models. *Extremes*, 29(2), 347-386.
- Alotaibi et al. (2025). Joint modeling of low and high extremes using a multivariate extended generalized Pareto distribution. *arXiv:2509.05982*.
- Richards et al. (2025). Modern extreme value statistics for Utopian extremes. *Extremes*, 28(1), 149-171.
- Rai et al. Fast parameter estimation of generalized extreme value distribution using neural networks. *Environmetrics* 35.3 (2024): e2845.

The amortisation gap

Even with infinite deployment speed, approximation error remains.

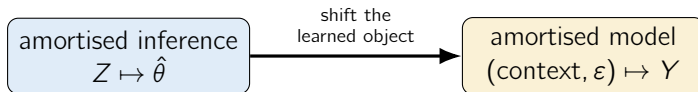
$$\underbrace{\text{total inferential error}}_{\text{what we care about}} \approx \underbrace{\text{statistical uncertainty}}_{\text{data}} + \underbrace{\text{approximation / optimisation error}}_{\text{network}} + \underbrace{\text{distribution shift}}_{\text{training vs reality}}.$$

Diagnostics should therefore be simulation-based

- test risk on independent simulated parameter/data pairs;
- empirical coverage of learned intervals;
- simulation-based calibration / posterior checks where appropriate;
- explicit stress tests under misspecification and out-of-distribution parameters.

Amortised generative models for cascading extremes

From estimating a model to learning a model



The simulator is fixed;
the inverse map is learned.

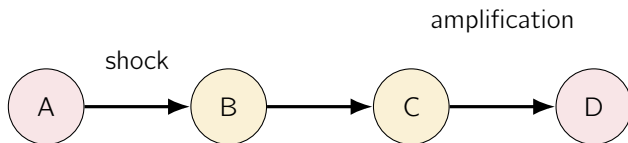
The conditional
distribution itself
is represented
by the network.

For cascading extremes, we want a reusable conditional generator for the temporal dependence of extreme-event tokens.

Cascading extremes

Extreme events often appear as **sequences of dependent shocks**, not isolated observations:

market crash, supply-chain disruption, natural hazards, power-grid failures.



Key target: $P(A_t \mid A_1, \dots, A_{t-1})$.

Turn a multivariate extreme process into a language

Let $\mathbf{Y}_t = (Y_{t,1}, \dots, Y_{t,d})^\top$ and $u_j = F_j^{-1}(q)$.

$$l_{t,j} = \mathbb{I}\{Y_{t,j} > u_j\}, \quad \mathcal{E}_t = \{j : l_{t,j} = 1\}, \quad \mathcal{V} = \{0, \dots, d\}.$$

$$A_t = \begin{cases} 0, & \mathcal{E}_t = \emptyset, \\ \arg \max_{j \in \mathcal{E}_t} F_j(Y_{t,j}), & \mathcal{E}_t \neq \emptyset. \end{cases}$$

- Token j identifies the **dominant exceedance on the marginal probability scale**; several components can exceed simultaneously.
- This many-to-one rule discards magnitudes and other simultaneous exceedances. Fix a deterministic tie rule for empirical scores.
- F is a rank-GPD model.

An “extremal sentence”

A cascade is a run of active tokens followed by a quiet token:

$$(A_s, \dots, A_e, 0), \quad A_s, \dots, A_e > 0.$$

The active run begins at the sequence boundary or after a quiet token.

$$0, 0, \underbrace{1, 4, 5, 0}_{\text{cascade}}, 0, \underbrace{2, 0}_{\text{single event}}, \underbrace{1, 3, 5, 0}_{\text{cascade}}, \dots$$

Quiet is both a state and a delimiter

Training retains the full token time series, including repeated zeros. A quiet token ends an individual cascade, but need not end generation of the time series.

Inferential target

Inferential target is the predictive distribution of a new token A_{t+1} , given the history $A_{\leq t}$:

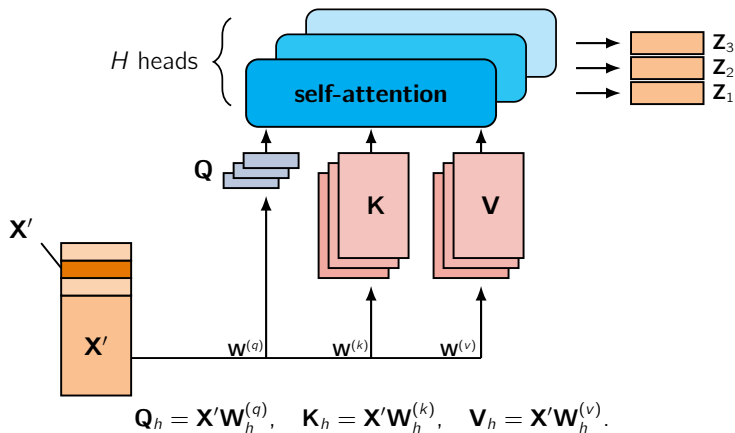
$$A_{t+1} \sim \text{Categorical}(\hat{\pi}_{\theta^*, t+1}). \quad (2)$$

The probability is parametrised by a transformer with parameters θ^* :

$$\hat{\pi}_{\theta^*, t+1} = \text{Transformer}_{\theta^*}(A_{\leq t}). \quad (3)$$

We seek to learn the θ^* that provides the optimal predictive distribution on **average** (over what??).

Multi-head self-attention



$$\mathbf{Z}_h = \text{Softmax}\left(\frac{\mathbf{Q}_h\mathbf{K}_h^\top}{\sqrt{d_{\text{emb}}/H}} + \mathbf{M}\right)\mathbf{V}_h.$$

Where does amortisation enter the generative model?

Draw DGP parameters and simulate a token sequence:

$$\psi^{(m)} \sim \Pi_{\psi}, \quad D^{(m)} \sim F(\cdot \mid \psi^{(m)}).$$

For $D = (A_1, \dots, A_N)$, minimise the next-token log loss:

$$\mathcal{L}(\theta; D) = -\frac{1}{N-1} \sum_{t=1}^{N-1} \log \hat{\pi}_{\theta, A_{t+1}}(A_{\leq t}).$$

$$\theta^* = \arg \min_{\theta} \mathbb{E}_{\psi \sim \Pi_{\psi}} \mathbb{E}_{D \sim F(\cdot \mid \psi)} [\mathcal{L}(\theta; D)].$$

- Refresh simulated regimes while continuing to update the same weights.
- The network sees tokens, not the true order or transition parameters.
- Log-loss training targets the task-mixture predictive law given the supplied history; DGP-specific recovery requires informative context.

Two families of simulated mechanisms

Heavy-tailed VAR family

$$\mathbf{Y}_t = \sum_{l=1}^p \mathbf{B}_l \mathbf{Y}_{t-l} + \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t = \text{diag}\{\boldsymbol{\sigma} \odot \exp(\mathbf{g}_t)\} \boldsymbol{\eta}_t, \quad \boldsymbol{\eta}_t \sim t_{5,\nu}(\mathbf{0}, \boldsymbol{\Sigma}).$$

Random coefficients, heavy tails and stochastic volatility; tokenize at $q = 0.98$, with thresholds estimated from an independent simulation.

Categorical family

A directly simulated p th-order token process with random lag-specific transition effects and an analytically known conditional law.

For both families, $\Pr(p = k) \propto 0.8^{k-1}$, $k = 1, \dots, 10$. **The latent VAR order is not generally the token-process order.**

Evaluate conditional recovery on unseen DGPs

- Independent training and test DGPs from the **same families and priors**; tests cover $p = 1, \dots, 10$.
- VAR reference: particle-filter approximation $\boldsymbol{\pi}^{\text{MC}}(A_{\leq t})$, with Monte Carlo error.

$$\text{KL}(\boldsymbol{\pi}, \hat{\boldsymbol{\pi}}) = \sum_{j=0}^5 \pi_j \log \frac{\pi_j}{\hat{\pi}_j}, \quad \text{TV}(\boldsymbol{\pi}, \hat{\boldsymbol{\pi}}) = \frac{1}{2} \sum_{j=0}^5 |\pi_j - \hat{\pi}_j|.$$

Compute errors for each history, then average; also compare active mass $1 - \pi_0$.

Scope of the experiment

Unseen DGP realizations within the training family demonstrate within-family generalisation, not arbitrary out-of-family extrapolation.

Does the learned generator recover next-event probabilities?

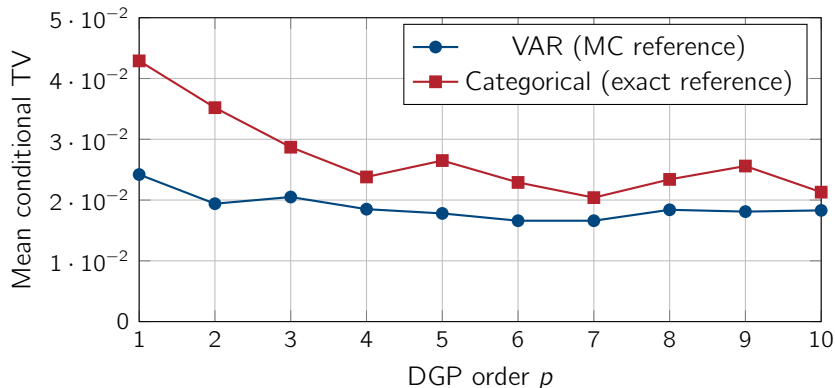
Family	Order p	Mean KL	Mean TV
VAR	1	0.0134	0.0242
VAR	5	0.0065	0.0178
VAR	10	0.0063	0.0183

Across all orders, mean TV ranges from 0.0166–0.0242 (VAR).

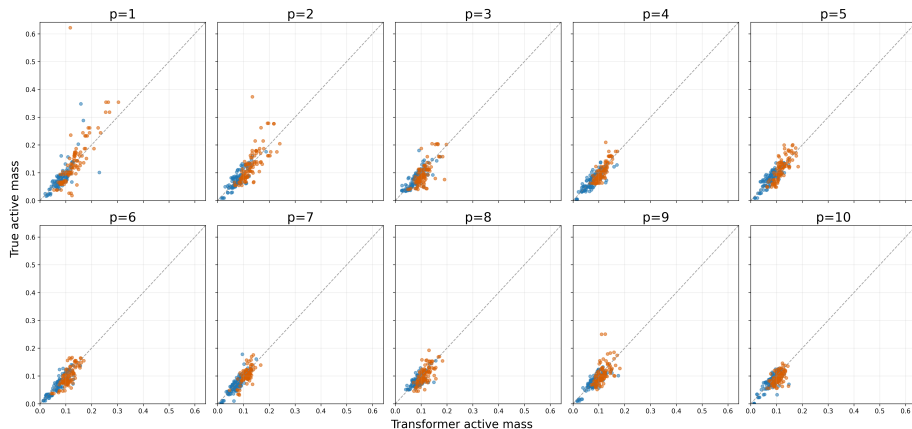
Interpretation

Small average conditional errors in these experiments. Lower errors at higher p do not establish that higher-order problems are intrinsically easier: the randomized, normalized lag effects also change task difficulty.

Conditional error across all ten orders



Conditional active-mass calibration



Transfer to semiconductor loss returns

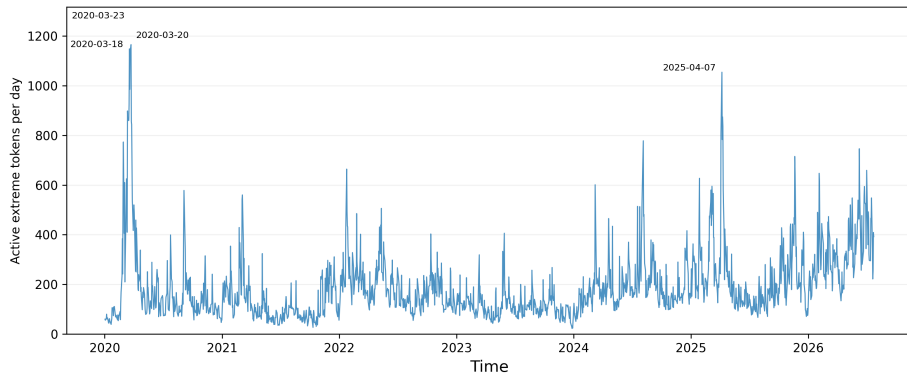
$$Y_{t,j} = -\log(P_{t,j}/P_{t-1,j}), \quad \mathcal{V} = \{\text{Quiet}, \text{NVDA}, \text{TSM}, \text{AVGO}, \text{MU}, \text{AMD}\}.$$

- Ten-second prices; larger positive returns mean larger price falls.
- Observation period: 2 January 2020–24 July 2026; tokenization at $q = 0.98$.
- Chronological split: train through March 2026, validate in April–May, test in June–24 July 2026.
- Forward-filled prices imply zero returns when unchanged; missing trading information should not be interpreted as direct evidence of low risk.

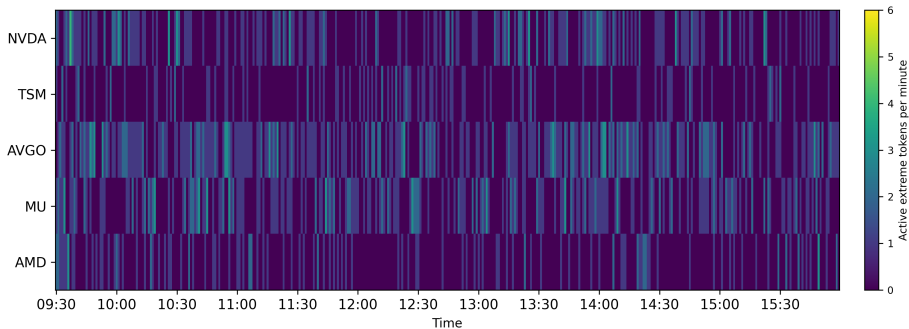
What is modelled?

The next dominant loss-exceedance token and its temporal dependence; neither loss magnitude nor a causal transmission mechanism.

Extreme tokens



Extreme tokens during the March 2020 crash



Adaptation and recursive generation

- **Zero-shot**: pre-trained weights, with no financial fine-tuning.
- **One-shot / few-shot**: early fine-tuning checkpoints; few-shot is five epochs in the paper. These labels refer to adaptation, not counts of individual examples.
- **“Optimal-shot”**: the checkpoint selected by validation cross-entropy, reached at epoch 18 among 200 epochs.
- Seed generation with the last 256 validation tokens, then append each sampled token and roll forward; 1,000 Monte Carlo repetitions.

Different evaluation targets

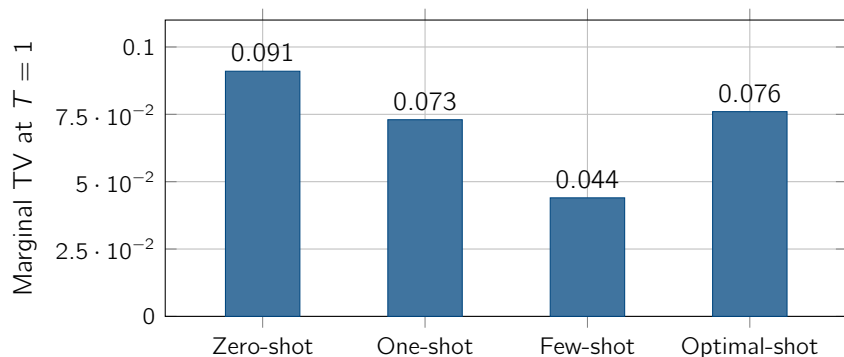
Validation log loss selects a predictive checkpoint. It need not select the best marginal or transition fit in long recursive rollouts.

Few-shot gives the best reported marginal fit

	Quiet	NVDA	TSM	AVGO	MU	AMD	TV
Observed	.727	.011	.056	.030	.101	.076	—
Zero-shot	.811	.016	.037	.032	.056	.048	.091
One-shot	.778	.024	.031	.039	.066	.063	.073
Few-shot	.756	.018	.040	.037	.084	.063	.044
Optimal-shot	.803	.009	.044	.026	.077	.041	.076

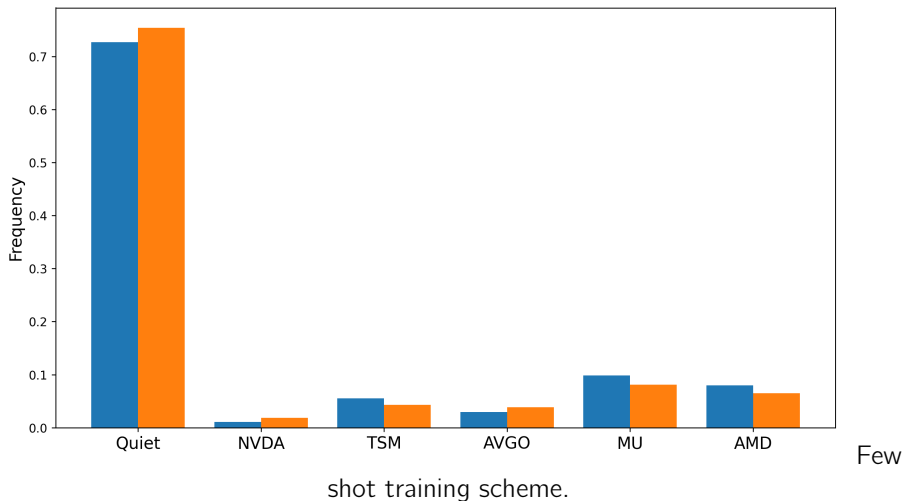
- At $T = 1$, few-shot reduces reported marginal TV from 0.091 to 0.044.
- Quiet remains overrepresented: 0.756 versus 0.727 observed.
- The largest few-shot active-token discrepancy in this rounded table is MU (0.017), followed by TSM (0.016).

Few-shot has the lowest reported rollout error

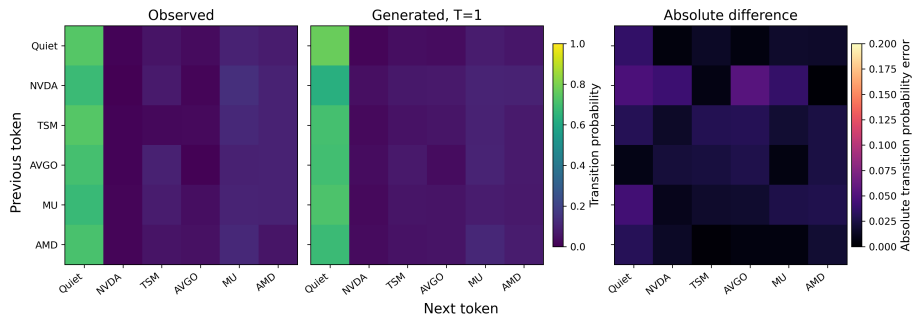


"Optimal-shot" is selected by validation cross-entropy, not by held-out marginal TV.

Observed and generated marginal token frequencies



Observed and generated one-step transitions



Few shot training scheme.

Inference and modelling: the parallel is now explicit

	Neural Bayes estimator	Cascading transformer
Task distribution	$\theta \sim \Omega, Z \sim p(\cdot \theta)$	$\psi \sim \pi_\psi, D \sim F(\cdot \psi)$
Learned map	$Z \mapsto \hat{\theta}$	history $A_{\leq t} \mapsto P(A_{t+1} A_{\leq t})$
Loss	Bayes decision loss	cross-entropy / log score
Deployment	parameter / posterior summary	prediction and recursive generation
Amortised over	datasets and parameter values	histories and cascading mechanisms

What makes a good amortised model for extremes?

- 1 **Tail-aware task distribution:** training must actually expose the network to the extreme regimes we care about.
- 2 **Structural constraints:** causality, permutation invariance, support constraints, marginal standardisation, censoring masks.
- 3 **Proper scoring / risk target:** train for the distributional object that matters, not only mean squared error.
- 4 **Out-of-distribution diagnostics:** stress-test new thresholds, dependence strengths, dimensions and mechanisms.
- 5 **Fast repeated use:** the downstream scientific workflow should genuinely benefit from reuse.

Some unresolved questions

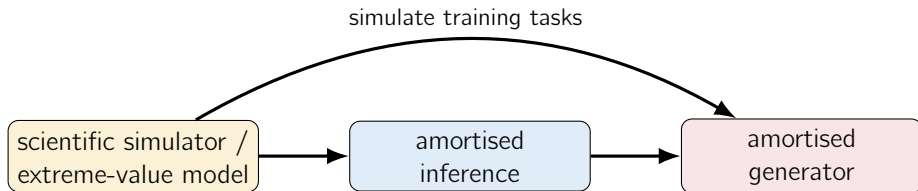
Statistical

- robustness to simulator misspecification;
- approximation theory;
- high-dimensional parameter sets
- extrapolation beyond the training prior/task distribution;
- identifiability of learned summaries and mechanisms.

Modelling

- continuous magnitudes as well as extreme indicators;
- simultaneous and delayed cascades;
- high-dimensional systems ($d \gg 100$);
- nonstationary covariates and interventions;
- interpretable causal pathways rather than black-box attention alone.

A broader research programme



Fast parameter learning, posterior summaries, emulation, conditional prediction and scenario generation can all be trained over deliberately chosen distributions of extreme-value tasks.

Take-home messages

- ➊ **Amortisation is a computational strategy:** pay a setup cost once, then exploit a cheap learned map many times.
- ➋ **Neural Bayes estimators** turn simulation from a model into fast Bayes decision rules, particularly useful when likelihoods are difficult but simulators are available.
- ➌ **Amortisation can be broadened** over thresholds, sampling designs, missingness patterns, parameter ranges and other task-level choices.
- ➍ The same philosophy extends from **inference** to **models**: train a reusable conditional generator across a family of extreme-event mechanisms.
- ➎ For extremes, success depends on **tail-aware training distributions, statistical structure and rigorous calibration**.

References I



Zammit-Mangion, A., Sainsbury-Dale, M. and Huser, R. (2025). Neural Methods for Amortized Inference. *Annual Review of Statistics and Its Application*, 12, 311–335.



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Sainsbury-Dale, M., Zammit-Mangion, A., Richards, J., and Huser, R. (2023/24). Neural Bayes estimators for irregular spatial data using graph neural networks. *JCGS*, 34(3), 1153–1168.

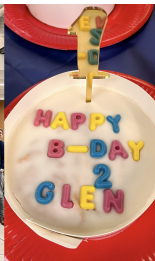


Huser, R. and Wadsworth, J. L. (2022). Advances in statistical modeling of spatial extremes. *WIREs Computational Statistics*, 14, e1537.

Thank you

Questions?

Some advertisements



02/10/26 14:30 BST Joint CIRCE/GLEN Seminar by Antonio Ocello (ENSAE Paris/CREST)

Do Heavy Tails Help Diffusion? On the Subtle Trade-off Between Initialization and Training <https://glen-scotland.github.io/glen/seminars.html>

Seminar series + Working Groups

- One World Extremes - <https://sites.google.com/view/ow-extremes/>
- Spatio-Temporal Statistics and Data Science Online Seminars - <https://www.stsds.org/index.html>
 - Sep. 25, 2026, 11:00 AM ET. *Modeling Spatio-temporal Extremes via Conditional Variational Autoencoders* by Likun Zhang
- TIES working groups HPCs, GenAI, Extremes - http://www.environmetrics.org/TIES_working_groups.html

EVA 2027

Montreal, July 5th–9th 2027.

<https://hecsciencesdecision.github.io/eva2027/>

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Data challenge

The data challenge for the Extreme Value Analysis 2027 conference is organized in collaboration with the [International Air Transport Association \(IATA\)](#), headquartered in Montréal. It consists of three prediction tasks related to the estimation of the extremal characteristics of in-flight turbulence events experienced by transatlantic flights.

Registration will open by the end of September 2026. Participants will need to fill a sign-up form and a non-disclosure agreement to access the data.

The contact person for the data challenge is [Jordan Richards](#) (University of Edinburgh)